Name: ____________________

STATISTICS, (CUMULATIVE FREQUENCY AND BOX PLOTS)

Date: ________________

LO: I can draw and interpret cumulative frequency diagrams and box plots.

Cumulative frequency and box plots

Cumulative frequency is the running total of frequencies. A **box plot** (box-and-whisker diagram) displays the minimum, lower quartile (Q1), median (Q2), upper quartile (Q3), and maximum.

Strategy: Add frequencies cumulatively, plot against upper class boundaries, then read off Q1 (at $\frac{n}{4}$), Q2 (at $\frac{n}{2}$), Q3 (at $\frac{3n}{4}$).

Quick examples

●	IQR = Q3 - Q1 measures the spread of middle 50%	interquartile range
●	Median is at the $\frac{n}{2}$ position on the CF curve	reading the median
●	Box plot shows min, Q1, Q2, Q3, max	five-number summary
●	Plot CF against midpoints	plot against upper boundaries

Key idea

Cumulative frequency = running total. Read quartiles from the CF graph. IQR = Q3 - Q1.

IQR = Q3 - Q1

The interquartile range measures spread of the middle 50% of data

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – building a CF table

Frequencies are 5, 8, 12, 10, 5. Build the cumulative frequency column.

CF: 5, 5+8=13, 13+12=25, 25+10=35, 35+5=40

Total n = 40

CF values: 5, 13, 25, 35, 40

5, 13, 25, 35, 40

Add each frequency to the running total.

Example 2 – reading the median

n = 40. Read the median from the CF graph.

Median position: $\frac{40}{2}$ = 20th value

Go across from CF = 20 to the curve

Read down to the x-axis for the median

Read at CF = 20

The median is the value at the $\frac{n}{2}$ position on the y-axis.

Example 3 – finding IQR

From a CF graph: Q1 = 35, Q3 = 58. Find the IQR.

Q1 is at $\frac{n}{4}$: value = 35

Q3 is at $\frac{3n}{4}$: value = 58

IQR = 58 - 35 = 23

IQR = 23

The IQR tells us the range of the middle 50% of the data.

Example 4 – drawing a box plot

Min = 12, Q1 = 25, Q2 = 33, Q3 = 45, Max = 60. Describe the box plot.

Draw a number line from 10 to 65

Box from Q1 (25) to Q3 (45), line at median (33)

Whiskers from 12 (min) to box and box to 60 (max)

Box: 25 to 45, median at 33, whiskers 12 and 60

Five values define the entire box plot.

YOUR TURN – PRACTICE (deliberate practice)

1. Building CF tables

Find the cumulative frequencies.

a) Frequencies: 3, 7, 10, 5, 5

b) Frequencies: 4, 6, 15, 8, 7

c) Frequencies: 2, 8, 12, 14, 4

d) Frequencies: 10, 15, 20, 10, 5

2. Finding quartile positions

State the CF positions for Q1, Q2, Q3.

a) n = 40

b) n = 60

c) n = 100

d) n = 80

3. Calculating IQR

Find the IQR from the given quartiles.

a) Q1 = 15, Q3 = 35

b) Q1 = 42, Q3 = 68

c) Q1 = 3.5, Q3 = 7.2

d) Q1 = 120, Q3 = 185

4. Box plot interpretation

From the box plot, find the stated value.

a) Min=10, Q1=20, Q2=35, Q3=50, Max=70. Range?

b) Same data. IQR?

c) Min=5, Q1=12, Q2=18, Q3=25, Max=40. Median?

d) Same data. What fraction is between Q1 and Q3?

5. Comparing distributions

Compare using the given statistics

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) IQR = Q3 - Q1 ☐

Correct answer: __________

Reason: __________________

b) CF should be plotted against class midpoints ☐

Correct answer: __________

Reason: __________________

c) The median is read at the $n/2$ position on the CF graph ☐

Correct answer: __________

Reason: __________________

d) The box in a box plot always shows 25% of the data ☐

Correct answer: __________

Reason: __________________

e) A box plot shows the five-number summary: min, Q1, Q2, Q3, max ☐

Correct answer: __________

Reason:

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A CF diagram shows n = 80 students. From the curve: Q1 = 42 marks, median = 55, Q3 = 67. The pass mark is 50. Estimate the number who passed. Draw a box plot if min = 20 and max = 90.

Expression: ____________________________

Simplified: ____________________________

b) Two box plots compare test scores. Class A: min 25, Q1 40, Q2 55, Q3 70, max 95. Class B: min 35, Q1 50, Q2 60, Q3 65, max 80. Write three comparisons using correct statistical language (median, IQR, range, skewness).

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!